Digital Manufacturing Blueprint & Roadmap (Illustrative)

Life Sciences | MES-Enabled Execution

Illustrative portfolio example (non-client specific)

Objective & Scope

Objective

- Improve manufacturing execution consistency, compliance, and visibility using MES-enabled digital workflows

Scope

- Electronic batch records (EBR) and execution workflows
- Equipment integration and automation interfaces
- Deviation handling and quality alignment
- Manufacturing data visibility and decision support

Observed Challenges in Regulated Manufacturing

- Workflow variability across manual or semi-digital execution
- Documentation effort and review time impacting batch cycle time
- Deviations require manual data collection across multiple systems
- Limited real-time visibility into batch status and execution progress
- Data fragmented across MES, historians, and spreadsheets

Target State

Target State Characteristics

- Standardized MES-driven electronic batch execution (EBR)
- System-enforced sequencing and procedural controls
- Improved integration between MES, control systems, and data layers
- Exception-based handling to support deviation readiness
- Better visibility into execution progress and performance

Target Outcomes

- Improved right-first-time execution
- Faster review readiness and improved throughput
- Stronger data integrity and traceability

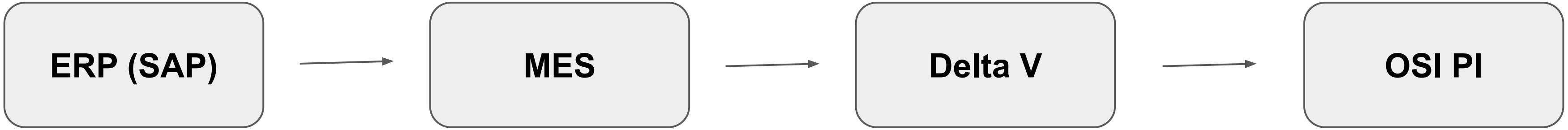
Digital Manufacturing Architecture

Digital Architecture Overview

- **Enterprise Layer:** ERP (SAP)
- **Execution Layer:** MES (Syncade / PAS-X)
- **Control Layer:** DeltaV / equipment HMIs
- **Data Layer:** OSI PI / reporting tools

Core MES Capabilities

- EBR authoring and execution
- Material and equipment tracking
- Exception handling + traceability
- Execution data capture for review readiness



Roadmap & Value

Phased Implementation Roadmap

- **Phase 1:** Foundation & standardization
- **Phase 2:** Execution & integration
- **Phase 3:** Optimization & analytics

Illustrative Value Case (from Artifact 2)

- Reduced deviation investigation effort
- Improved cycle time and batch review readiness
- Lower manual documentation burden
- Improved right-first-time execution (RFT)
- Better decision support through improved visibility

Impacts are illustrative and depend on scope, maturity, adoption, and validation constraints.